

Week7

Question1

Correct

Marked out of 1.00

Question text

Sunny and Johnny like to pool their money and go to the ice cream parlour. Johnny never buys the same flavour that Sunny does. The only other rule they have is that they spend all of their money.

Given a list of prices for the flavours of ice cream, select the two that will cost all of the money they have.

For example, they have $m=6$ to spend and there are flavours costing $\text{cost}=[1,2,3,4,5,6]$. The two flavours costing 1 and 5 meet the criteria. Using 1-based indexing, they are at indices 1 and 4. 1 and 4.

Function Description

Complete the code in the editor below. It should return an array containing the indices of the prices of the two flavours they buy

It has the following:

- m : an integer denoting the amount of money they have to spend
- cost : an integer array denoting the cost of each flavour of ice cream

Input Format

The first line contains an integer, t , denoting the number of trips to the ice cream parlour. The next t set of lines each describe a visit. Each trip is described as follows:

1. The integer m , the amount of money they have pooled.
2. The integer n , the number of flavours offered at the time.
3. n space-separated integers denoting the cost of each flavour: $\text{cost}[\text{cost}[1], \text{cost}[2], \dots, \text{cost}[n]]$. The

Note: index within the cost array represents the flavour of the ice cream purchased.

Constraints

$$1 \leq t \leq 50$$

$$2 \leq m \leq 10^4$$

$$2 \leq n \leq 10^4$$

$$1 \leq \text{cost}[i] \leq 10^4, \forall i \in [1, n]$$

There will always be a unique solution

Output Format

Foreachtestcase,printtwospace-separatedintegersdenotingtheindicesofthetwoflavourspurchased, in ascending order.

Sample Input

2

4

5

14532

4

4

2243

Sample Output

t1

4

12

Explanation

Sunny and Johnny make the following two trips to the parlor:

1. The first time, they pool together $m=4$ dollars. Of the five flavors available that day, flavors 1 and 4 have a total cost of $1 + 3 = 4$.

2. The second time, they pool together $m=4$ dollars. Of the four flavors available that day, flavors 1 and 2 have a total cost of $2 + 2 = 4$.

```

1 #include<stdio.h>
2 int main()
3 {
4     int t,m,n,c=0;
5     scanf("%d",&t);
6     for(int i=0;i<t;i++)
7     {
8         c=0;
9         scanf("%d\n%d",&m,&n);
10        int arr[n];
11        for(int j=0;j<n;j++)
12        {
13            scanf("%d",&arr[j]);
14        }
15        for(int a=0;a<n-1;a++)
16        {
17            for(int b=a+1;b<n;b++)
18            {
19                if(arr[a]+arr[b]==m)
20                {
21                    printf("%d %d\n",a+1,b+1);
22                    c=1;break;
23                }
24                if(c==1)break;
25            }
26        }
27    }
28    return 0;
29 }

```

	Input	Expected	Got	
✓	2	1 4	1 4	✓
	4	1 2	1 2	
	5			
	1 4 5 3 2			
	4			
	4			
	2 2 4 3			

Passed all tests! ✓

Question2

CorrectMarkedoutof1.00 Question

text

NumerostheArtisthadtwolists thatwerepermutationsofoneanother.Hewasveryproud. Unfortunately,whiletransportingthemfromoneexhibitiontoanother,somenumberswerelostoutofthe first list. Can you find the missing numbers?

Asanexample,thearraywithsomenumbersmissing,arr=[7,2,5,3,5,3].Theoriginalarrayofnumbers brr = [7, 2, 5, 4, 6, 3, 5, 3]. The numbers missing are [4, 6].

Notes

- Ifanumberoccursmultipletimesinthelists,youmustensurethatthefrequencyofthatnumberinboth lists is the same. If that is not the case, then it is alsoa missing number.
- Youhavetoprintallthemissingnumbersinascendingorder.
- Printeachmissingnumberonce,evenifitissingingmultipletimes.

The difference between maximum and minimum number in the second list is less than or equal to 100.

complete the code in the editor below. It should return an array of missing numbers. It has the following:

- arr: the array with missing numbers
- brr: the original array of numbers

Input Format

There will be four lines of input:

n - the size of the first list, arr

The next line contains n space-separated integers arr[i]

m - the size of the second list, brr

The next line contains m space-separated integers brr[i] Constraints

- $1 \leq n, m \leq 2 \times 10^5$
- $n \leq m$
- $1 \leq brr[i] \leq 2 \times 10^4$
- $X_{\max} - X_{\min} < 101$

Output Format

Output the missing numbers in ascending order

Sample Input

10

203204205206207208203204205206

13

203204204205206207205208203206205206204

Sample Output

204 205 206

Explanation 204 is present in both arrays. Its frequency in arr is 2, while its frequency in brr is 3. Similarly, 205 and 206 occur twice in arr, but three times in brr. The rest of the numbers have the same frequencies in both lists.

Answer: (penalty regime: 0%)

```

1 #include<stdio.h>
2 int main()
3 {
4     int n,m,c,c1=0,co;
5     scanf("%d",&n);
6     int arr[n];
7     for(int a=0;a<n;a++)
8     {
9         scanf("%d",&arr[a]);
10    }
11    scanf("%d",&m);
12    int brr[m],ans[m];
13    for(int b=0;b<m;b++)
14    {
15        scanf("%d",&brr[b]);
16    }
17    for(int j=0;j<m;j++)
18    {
19        c=0;
20        for(int i=0;i<n;i++)
21        {
22            if(arr[i]==brr[j])
23            {
24                c=1;
25                arr[i]=-1;
26                break;
27            }
28        }
29        if(c==0)
30        {
31            ans[c1]=brr[j];
32            c1++;
33        }
34    }

```

```

33    }
34    }
35    for(int a=0;a<c1;a++)
36    {
37        co=0;
38        for(int b=0;b<c1;b++)
39        {
40            if(ans[b]<ans[a])
41                co++;
42        }
43        int temp=ans[a];
44        ans[a]=ans[co];
45        ans[co]=temp;
46    }
47    for(int i=0;i<c1;i++)
48        printf("%d ",ans[i]);
49    return 0;
50 }

```

	Input	Expected	Got	
✓	10 203 204 205 206 207 208 203 204 205 206 13 203 204 204 205 206 207 205 208 203 206 205 206 204	204 205 206	204 205 206	✓

Passed all tests! ✓

3,

Question text

Watson gives Sherlock an array of integers. His challenge is to find an element of the array such that the sum of all elements to the left is equal to the sum of all elements to the right. For instance, given the array **arr = [5, 6, 8, 11]**, **8** is between two subarrays that sum to **11**. If your starting array is **[11]**, that element satisfies the rule as left and right sum to **0**.

You will be given arrays of integers and must determine whether there is an element that meets the criterion.

Complete the code in the editor below. It should return a string, either YES if there is an element meeting the criterion or NO otherwise.

It has the following:

- `arr`: an array of integers

72

Input Format

The first line contains **T**, the number of test cases.

The next **T** pairs of lines each represent a test case.

- The first line contains **n**, the number of elements in the array **arr**.
- The second line contains **n** space-separated integers **arr[i]** where $0 \leq i < n$.

Constraints

- $1 \leq T \leq 10$
- $1 \leq n \leq 10^5$

Constraints

- $1 \leq T \leq 10$
- $1 \leq n \leq 10^5$
- $1 \leq arr[i] \leq 2 \times 10^4$
- $0 \leq i \leq n$

Output Format

For each test case print YES if there exists an element in the array, such that the sum of the elements on its left is equal to the sum of the elements on its right; otherwise print NO.

Sample Input 0

```
2
3
1 2 3
4
1 2 3 3
```

Sample Output 0

```
NO
```

```
YES
```

Explanation 0

For the first test case, no such index exists.

For the second test case, $arr[0] + arr[1] = arr[3]$, therefore index **2** satisfies the given conditions.

Sample Input 1

73

```
3
5
1 1 4 1 1
4
2 0 0 0
4
0 0 2 0
```

Sample Output 1

```
YES
YES
YES
```

Explanation 1

In the first test case, **arr[2] = 4** is between two subarrays summing to **2**.

In the second case, **arr[0] = 2** is between two subarrays summing to **0**.

In the third case, **arr[2] = 2** is between two subarrays summing to **0**.

```

1 #include<stdio.h>
2 int main()
3 {
4     int t,n,Is,rs,m;
5     scanf("%d",&t);
6     for(int i=0;i<t;i++)
7     {
8         Is=0;
9         rs=0;
10        scanf("%d",&n);
11        int arr[n];
12        for(int j=0;j<n;j++)
13            scanf("%d",&arr[j]);
14        m=n/2;
15        if(arr[m]==0)
16        {
17            for(m=0;arr[m]==0 && m<n;m++);
18        }
19        for(int j=0;j<=m;j++)
20            Is=Is+arr[j];
21        for(int j=m;j<n;j++)
22            rs=rs+arr[j];
23        printf("%s\n",(Is==rs)?"YES" : "NO");
24    }
25    return 0;
26 }

```

	Input	Expected	Got	
✓	3	YES	YES	✓
	5	YES	YES	
	1 1 4 1 1	YES	YES	
	4			
	2 0 0 0			
	4			
	0 0 2 0			
✓	2	NO	NO	✓
	3	YES	YES	
	1 2 3			
	4			
	1 2 3 3			

Passed all tests! ✓